

Coding interview questions

Swapping 2 numbers – with extra variable

swapping two values with extra variable

```
x = 10
y = 20
print('x = ',x)
print('y = ',y)

temp = x
x = y
y = temp

print('after swapping..')
print('x = ',x)
print('y = ',y)
```

Swapping 2 numbers – without extra variable

swapping two values without extra variable

```
x = 10
```

```
y = 20
```

```
print('x = ',x)
```

```
print('y = ',y)
```

```
x = x+y
```

```
y = x-y
```

```
x = x-y
```

```
print('after swapping..')
```

```
print('x = ',x)
```

```
print('y = ',y)
```

Factorial program

```
#ask user to enter a number to find factorial of it  
num = int(input('enter a number : '))  
res = 1 #to store result  
  
for i in range(num,1,-1):  
    res = res*i  
  
print('factorial of ',num,'is',res)
```

Fibonacci series

```
fno = 0
sno = 1
print(fno)
print(sno)
res = fno+sno
```

#assume that I want to print fibonacci series upto 100

```
while res<100:
    print(res)
    fno = sno
    sno = res
    res = fno+sno
```

Prime number program

```
num = int(input('enter a number to find prime or not : '))
isprime = True

for i in range(2,num):
    if num%i==0:
        isprime = False
        break

if isprime==True:
    print(num,'is prime number')
else:
    print(num,'is not prime number')
```

Palindrome number

```
num = int(input('enter a number to find palindrome or not : '))
temp = num #store original number into temp
rev = 0 #is used to store reversed number

while num>0:
    last = num%10 #we will get last digit
    num = num//10 #we will get all digits except last digit
    rev = (rev*10)+last

print('reversed numbers is : ', rev)

if rev==temp:
    print(temp,'is palindrome number')
else:
    print(temp,'is not palindrome number')
```

Armstrong number

```
num = int(input('enter a number to find armstrong or not : '))
temp = num #store original number into temp
result = 0 #is used to store result number

while num>0:
    last = num%10 #we will get last digit
    num = num//10 #we will get all digits except last digit
    result = result + (last*last*last)

print('sum of cubes is : ', result)

if result==temp:
    print(temp,'is armstrong number')
else:
    print(temp,'is not armstrong number')
```


Biggest element in a list

```
mylist = [10,9,11,23,8]
big = mylist[0] #assume that 0th element is biggest

#now read each element from the list
for item in mylist:
    #compare each element with big
    if big < item:
        big = item

print('biggest is..', big)
```

Searching an element in an array

```
mylist = [10,8,9,11,12]
element = 11 #assume that I want to find position of 11
pos = -1 #assume that position is -1 which means element not found

#now read each element from the list based on position
for i in range(len(mylist)):
    #compare element with each element of list
    if element==mylist[i]:
        pos = i
        break #go out of loop once you find the position

if pos== -1:
    print(element,'not found')
else:
    print(element,'found at position',pos)
```

Sorting an Array – bubble sort

```
mylist = [10,8,-9,11,12]
len = len(mylist)

for i in range(len):
    for j in range(len-i-1):
        if mylist[j]>mylist[j+1]:
            mylist[j], mylist[j+1] = mylist[j+1],mylist[j]

print(mylist)
```

Removing duplicates from array

```
mylist = [10,8,10,8,12]
len = len(mylist)

for i in range(len-1):
    for j in range(i+1,len-1):
        if mylist[i]==mylist[j] and mylist[i]!=-1:
            #replace duplicate element with -1
            mylist[j] = -1

for item in mylist:
    #print all elements except -1
    if item!=-1:
        print(item)
```

Reverse a string

```
name = 'palle'  
len = len(name)  
  
for i in range(len-1, -1, -1):  
    print(name[i], end='')
```

Palindrome string program

```
name = 'liril'  
rev = ''  
len = len(name)  
  
for i in range(len-1,-1,-1):  
    rev = rev+name[i]  
  
print('original string is', name)  
print('reversed string is', rev)  
  
if rev==name:  
    print('its is palindrome string')  
else:  
    print('it is not palindrome string')
```

Counting words in a String

```
name = 'palle technologies python data science'  
count = 1
```

```
for item in name:  
    if item == ' ':  
        count = count + 1
```

```
print('total words are..', count)
```


Reversing words in a String

```
name = 'palle technologies python data science'
#split into words based on spaces and store in list
mylist = name.split(' ')

#now read each word from list
for item in mylist:
    length = len(item)
    #now reverse that word
    for i in range(length-1,-1,-1):
        print(item[i],end='')
    print(' ',end='') #print space between words
```


Count vowels in a String

```
name = 'palle technologies'
count = 0

for item in name:
    if item in ('a','e','i','o','u'):
        count = count + 1

print('total vowels are..',count)
```

Right angle triangle pattern

```
for i in range(1,5):  
    for j in range(1,i+1):  
        print('*',end=' ')  
    print()
```

```
*  
**  
***  
****
```

Reversed right angle triangle pattern

```
for i in range(4,0,-1):  
    for j in range(1,i+1):  
        print('*',end=' ')  
    print()
```



Mirrored right angle triangle pattern

```
for i in range(1,5):  
    for j in range(1,5):  
        if i+j<=4:  
            print(' ',end='')  
        else:  
            print('*',end='')  
    print()
```

```
      *  
     **  
    ***  
   ****
```

Pyramid star patter

```
for i in range(1,5):  
    for j in range(1,5+i-1):  
        if i+j<=4:  
            print(' ',end=' ')  
        else:  
            print('*',end=' ')  
    print()
```

